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DIPLOMA

3rd SEMESTER

COMPUTER NETWORKS

Unit-01 Lec-01

- COMPLETE SYLLABUS
- EASY EXPLANATION
- EXAM PATTERN
- IMPORTANT TOPICS

NEW

CS - 3RD SEMESTER

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UNIT 1: INTRODUCTION :

(06 Periods)

- Introduction to computer networks;
- Network Models- OSI Reference Model, TCP/IP Model;

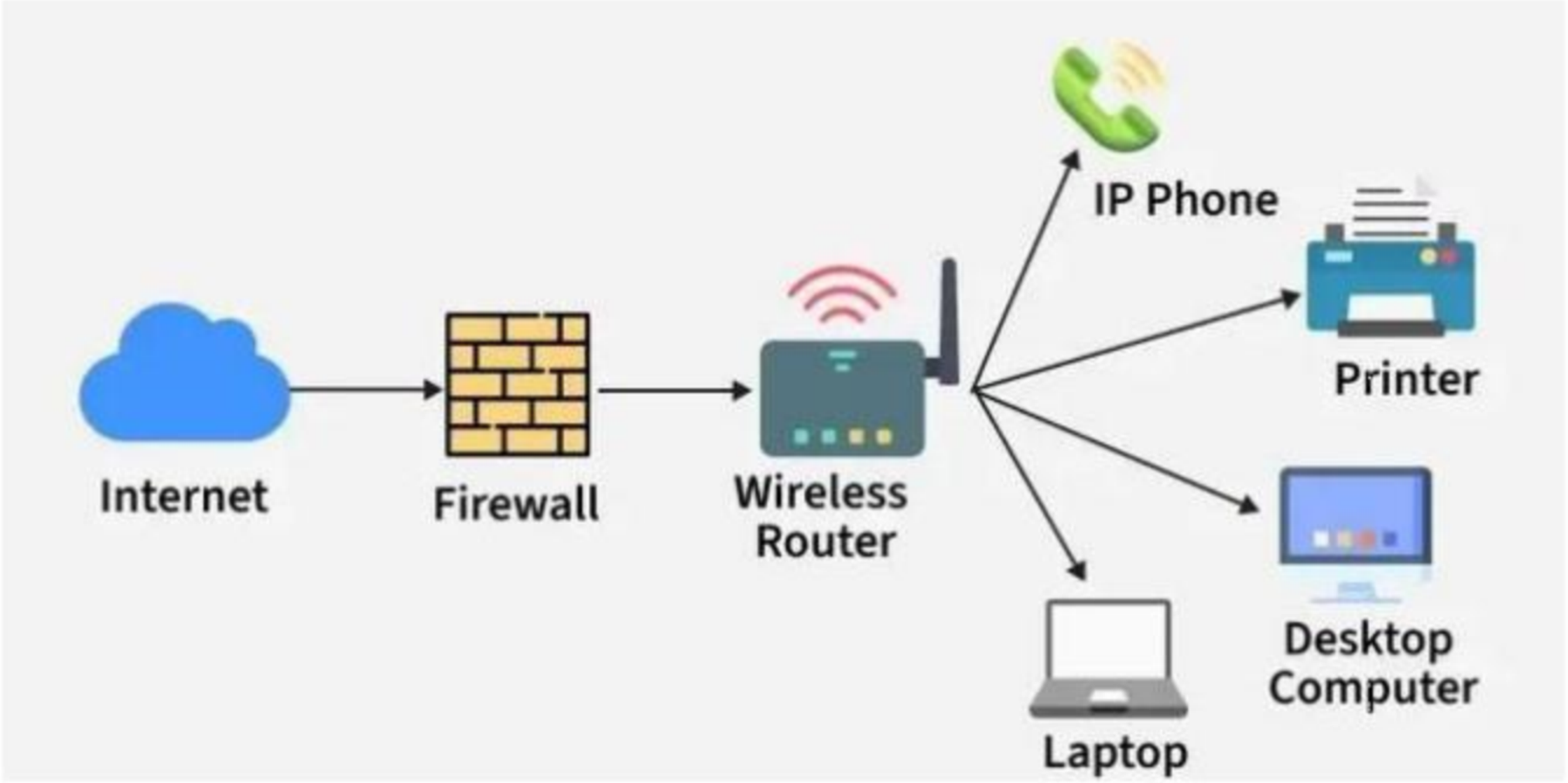
NETWORK • SERVER • DATA TRANSFER • SECURITY

Introduction to Computer Network

A computer network is a group of two or more computers that are connected to share resources (like files, printers, internet) and communicate with each other.

Computer Network दो या दो से अधिक computers का एक समूह है जो resources (जैसे files, printers, Internet) को share करने और एक-दूसरे के साथ communication करने के लिए connected होते हैं।

Wired / Wireless



Intra Communication

- **Intra Communication refers to communication that takes place within the same organization, network, department, or system.**
- **Intra Communication वह communication है जो एक ही organization, network, department या system के अंदर होती है।**

- **The sender and receiver belong to the same entity or internal environment.**
  - **इसमें sender और receiver एक ही entity या internal environment का हिस्सा होते हैं।**

- **Example: Communication between employees of the same company through the company network.**
- **Company network के माध्यम से एक ही company के employees के बीच communication।**

Inter Communication

- Inter Communication refers to communication between different organizations, networks, systems, or groups.
  - Inter Communication वह communication है जो अलग-अलग organizations, networks, systems या groups के बीच होती है।

- The sender and receiver belong to different entities.
- इसमें sender और receiver अलग-अलग entities से संबंधित होते हैं।

Delhi

Kerala

- **Example: Sending an email from one company to another company through the Internet.**
  - **Internet के माध्यम से एक company से दूसरी company को email भेजना।**

Main Purpose (मुख्य उद्देश्य)

- To share data and information.
  - Data और information को share करने के लिए।

- To share hardware devices (like printers, scanners).
  - Hardware devices (जैसे printers, scanners) को share करने के लिए।

- To allow communication (emails, chats, video calls).
  - Communication (emails, chats, video calls) की सुविधा प्रदान करने के लिए।

- To save time and money.
  - समय और पैसे की बचत करने के लिए।

Development of Computer Network

1. 1950s: Standalone Computers

- Computers worked independently and were not connected to each other.

  Computers स्वतंत्र रूप से काम करते थे और एक-दूसरे से connected नहीं थे।

- Data sharing was done manually using punched cards and magnetic tapes.

  Data sharing manually, punched cards और magnetic tapes की मदद से की जाती थी।

- Communication between computers was very difficult and time consuming.

  Computers के बीच communication करना बहुत कठिन और समय लेने वाला था।

2. 1960s: Introduction of Packet Switching

- Researchers developed the concept of packet switching.
- Researchers ने packet switching की concept विकसित की।

- Data was divided into small packets before transmission.

  Transmission से पहले data को छोटे-छोटे packets में divide किया जाता था।

- This technology became the foundation of modern networking.

  यह technology आधुनिक networking की नींव बनी।

3. 1969: ARPANET

- ARPANET was developed by the U.S. Department of Defense.

  ARPANET को U.S. Department of Defense द्वारा विकसित किया गया।

- It connected four universities through a packet-switched network.

  इसने चार universities को packet-switched network के माध्यम से जोड़ा।

- ARPANET is considered the first computer network and the predecessor of the Internet.

  ARPANET को पहला computer network और Internet का predecessor (पूर्वज) माना जाता है।

4. 1970s: Development of TCP/IP

- Vinton Cerf and Robert Kahn developed TCP/IP protocols.
- **Vinton Cerf** और **Robert Kahn** ने **TCP/IP** protocols विकसित किए।

- These protocols allowed different networks to communicate with each other.
- इन protocols ने अलग-अलग networks को आपस में communication करने योग्य बनाया।

- TCP/IP became the standard protocol suite for networking.
- TCP/IP, networking का standard protocol suite बन गया।

5. 1983: Birth of the Internet

- ARPANET officially adopted TCP/IP.
  - ARPANET ने आधिकारिक तौर पर TCP/IP को अपनाया।

- Different networks could now connect and exchange information.
  - अब अलग-अलग networks एक-दूसरे से connect होकर information का आदान-प्रदान कर सकते थे।

- This event is considered the birth of the modern Internet.
  - इस घटना को आधुनिक Internet की शुरुआत (Birth of the Internet) माना जाता है।

6. 1980s: Growth of LAN

- Local Area Networks (LANs) became popular in schools, colleges and offices.
  - **Local Area Networks** (LANs) स्कूलों, कॉलेजों और कार्यालयों में लोकप्रिय हो गए।

- Ethernet technology enabled faster communication.
  - Ethernet technology ने communication को अधिक तेज़ बनाया।

- Resource sharing became easier and more efficient.
  - Resource sharing अधिक आसान और प्रभावी (efficient) हो गई।

7.1990s: World Wide Web (WWW)

- Tim Berners-Lee introduced the World Wide Web.
  - Tim Berners-Lee ने World Wide Web (WWW) का विकास किया।

- Web browsers made Internet access simple for common users.
  - Web browsers ने सामान्य users के लिए Internet का उपयोग आसान बना दिया।

- The Internet expanded rapidly across the world.
  - Internet का पूरे विश्व में तेजी से विस्तार हुआ।

8. 2000s to Present

- Broadband Internet and Wi-Fi became common.
  - Broadband Internet और Wi-Fi का व्यापक रूप से उपयोग होने लगा।

- Mobile networks evolved from 2G to 5G.
  - Mobile networks 2G से 5G तक विकसित हुए।

- Cloud computing, IoT and AI-based networking emerged.
  - Cloud Computing, IoT (Internet of Things) और AI-based networking technologies का विकास हुआ।

Basic Components of a Network

- Nodes (computers, printers, etc.)
- Links (cables, wireless signals)
- Network devices (routers, switches, hubs)

Advantages of Computer Networks

- **Easy data sharing**

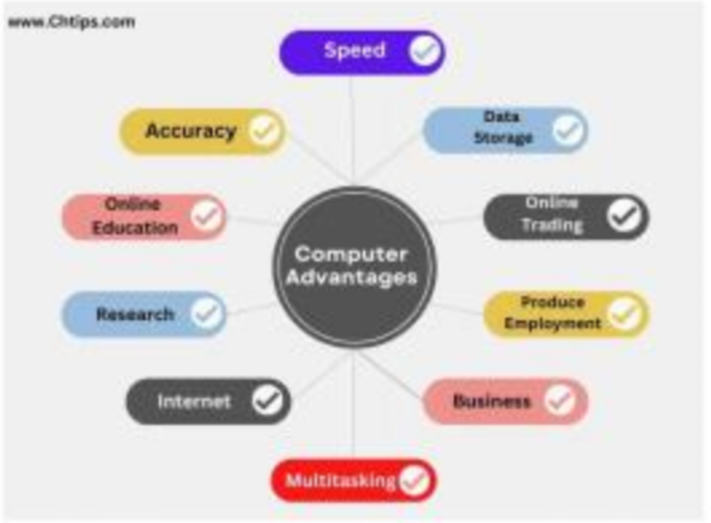
  Data को आसानी से share किया जा सकता है।

- **Cost-efficient resource sharing**

  Resources को share करके लागत (cost) कम की जा सकती है।

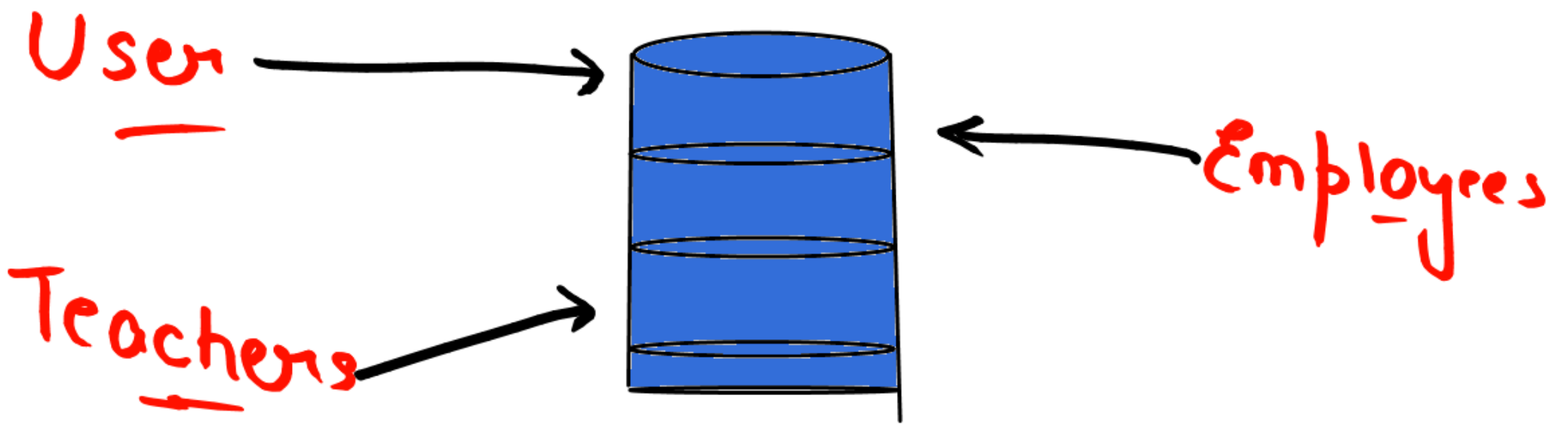
- **Better communication**

  बेहतर और तेज communication की सुविधा मिलती है।



- **Centralized data management**

  Data को एक central location से manage किया जा सकता है।



Examples in Daily Life

- Using Wi-Fi at home
  - घर पर Wi-Fi का उपयोग करना

- ATM network for banking
  - Banking के लिए ATM Network का उपयोग करना।

- Online classes and cloud storage
  - Online classes और Cloud Storage का उपयोग करना।

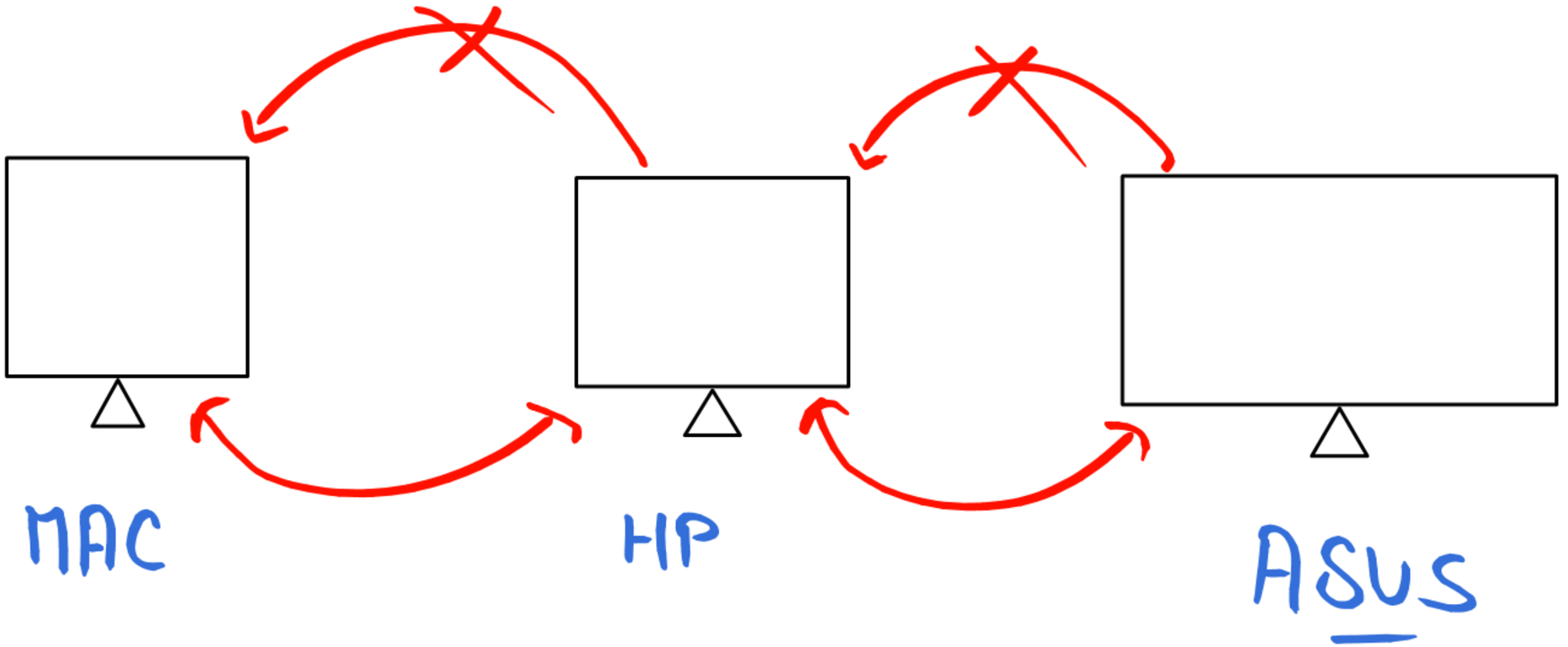
NETWORK MODEL

A Network Model is a framework that describes how data is transmitted from one device to another in a computer network.

Network Model एक framework है जो बताता है कि computer network में data एक device से दूसरी device तक कैसे transmit होता है।

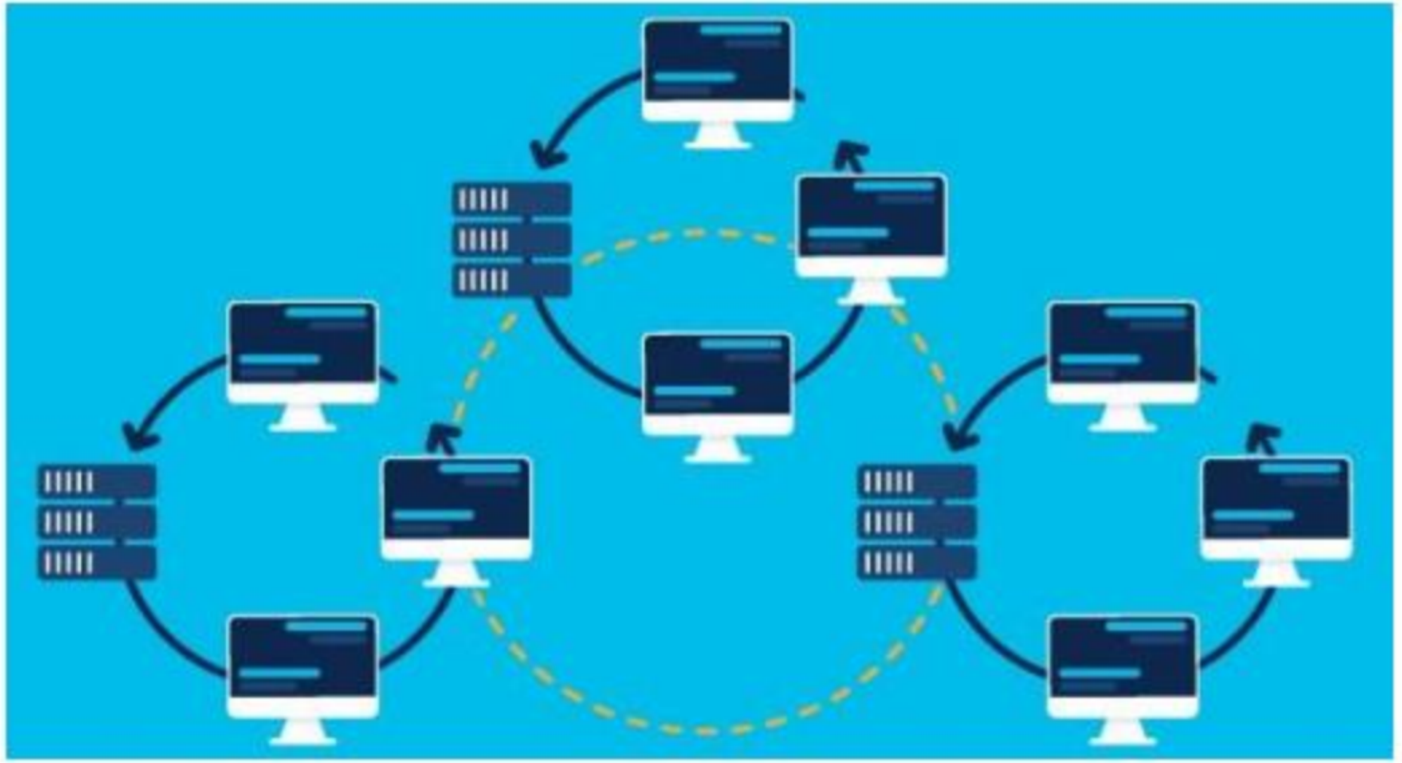
It divides the communication process into different layers, where each layer performs a specific function.

संचार प्रक्रिया को विभिन्न परतों में विभाजित करता है, जहां प्रत्येक परत एक विशिष्ट कार्य करती है।


MAC
HP
ASUS

Example: The OSI Model and TCP/IP Model are the most common network models used in computer networking.

उदाहरण: OSI Model और TCP/IP Model, computer networking में सबसे अधिक उपयोग किए जाने वाले network models हैं।



Protocol

A Protocol is a set of rules and standards that govern communication between devices in a network.

Protocol नियमों (rules) और standards का एक समूह है जो network में devices के बीच communication को नियंत्रित करता है।

- It defines how data is transmitted, received, and interpreted.

यह निर्धारित करता है कि data को कैसे transmit, receive और interpret किया जाएगा।

Example:

- HTTP (Hypertext Transfer Protocol)
- HTTPS (Secure HTTP)
- FTP (File Transfer Protocol)
- SMTP (Simple Mail Transfer Protocol)
- TCP (Transmission Control Protocol)

/ POP 3

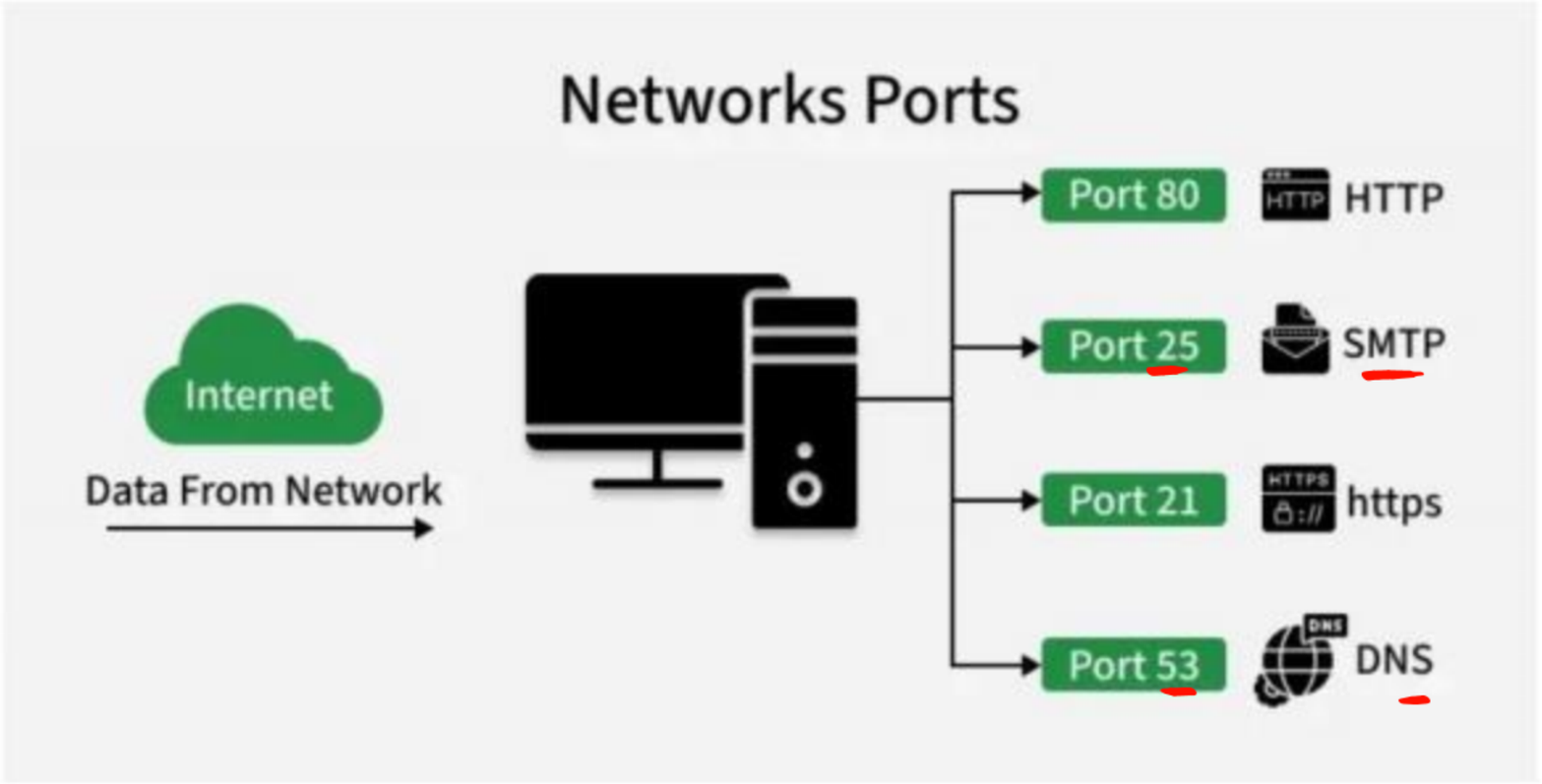
Port

- **A Port is a logical communication endpoint used by applications and services in a network.**
  - **Port एक logical communication endpoint है जिसका उपयोग network में applications और services द्वारा किया जाता है।**

- **It helps identify a specific process or service running on a device.**
  - **यह device पर चल रही किसी specific process या service की पहचान करने में मदद करता है।**

- **Example:** Web servers use Port 80 for HTTP and Port 443 for HTTPS.
- **उदाहरण:** Web servers, HTTP के लिए Port 80 और HTTPS के लिए Port 443 का उपयोग करते हैं।

HTTP/80 HTTPS/443



OSI MODEL (OPEN SYSTEM INTERCONNECTION MODEL)

- **The OSI (Open Systems Interconnection) Model is a reference model developed by ISO (International Organization for Standardization) to standardize network communication.**
  - **OSI (Open Systems Interconnection) Model एक reference model है जिसे ISO (International Organization for Standardization) द्वारा network communication को standardize करने के लिए विकसित किया गया।**

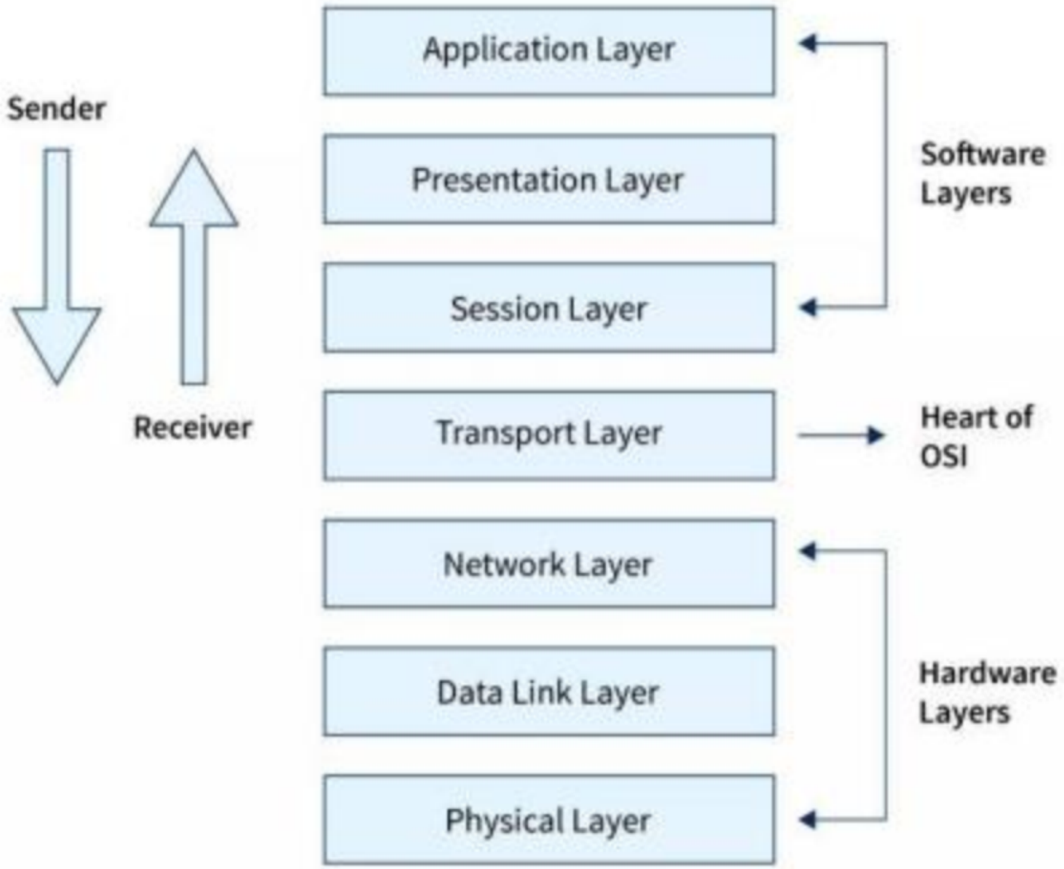
- It divides the communication process into seven layers, where each layer performs a specific function.
  - यह communication process को 7 layers में divide करता है, जहाँ प्रत्येक layer एक specific function करती है।

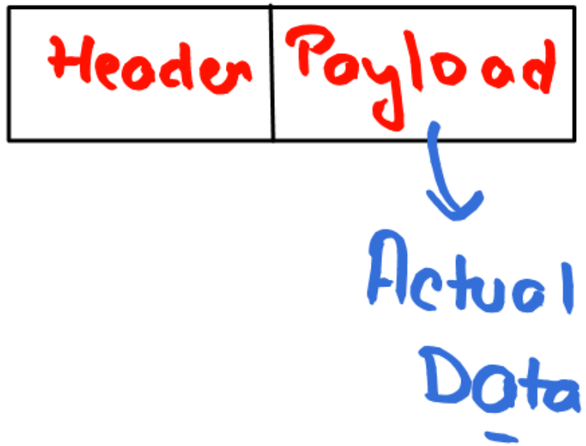
- The OSI model helps in understanding, designing, and troubleshooting computer networks.
  - OSI Model computer networks को understand, design और troubleshoot करने में मदद करता है।

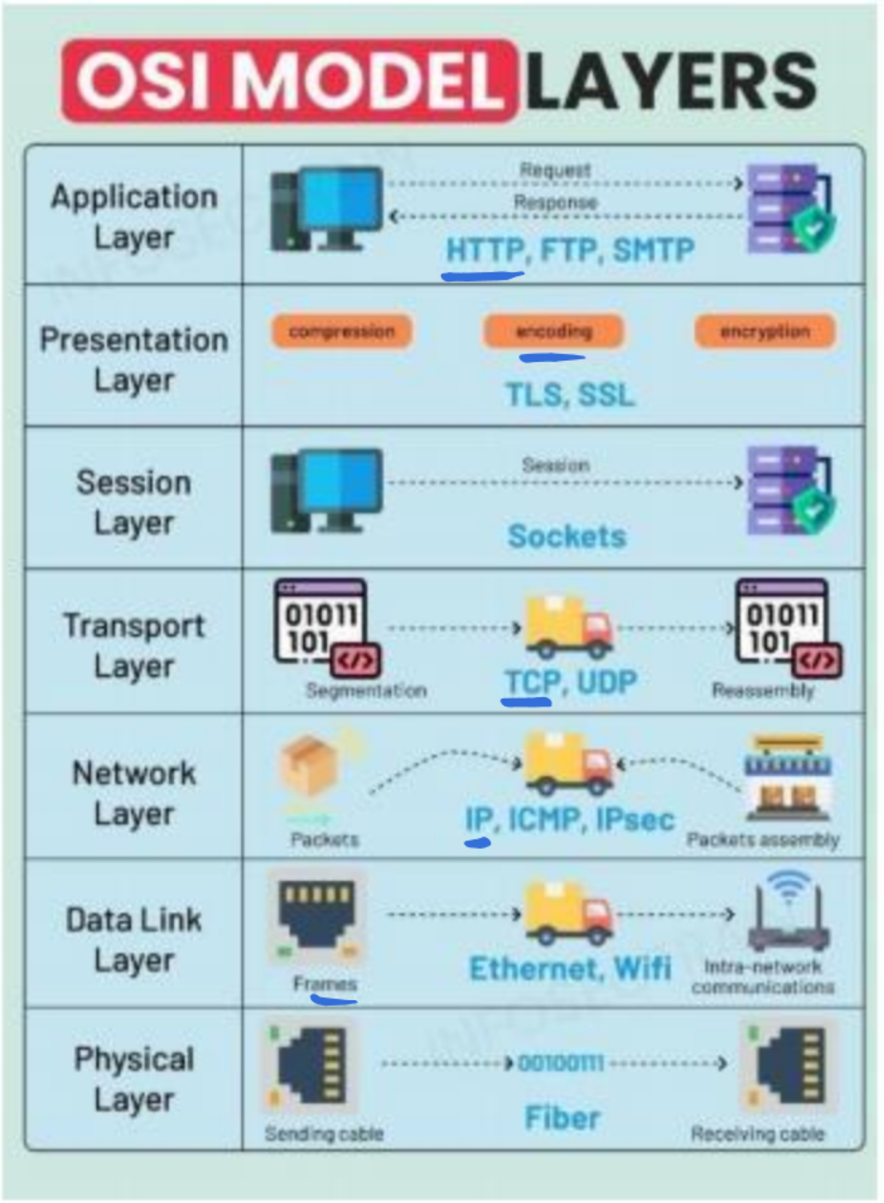
- OSI Model is a conceptual framework used to understand how data travels from one computer to another over a network.
  - OSI Model एक conceptual framework है जो यह समझने में मदद करता है कि data एक computer से दूसरे computer तक network के माध्यम से कैसे travel करता है।

- Each layer performs a specific function and provides services to the layer above it.
  - प्रत्येक layer एक specific function करती है और अपने ऊपर वाली layer को services प्रदान करती है।

Aaj
Phir
Se
Test
Nhi
Dena
Padega




Header
Payload
Actual Data


OSI MODEL LAYERS
Application Layer
Request
Response
HTTP, FTP, SMTP
Presentation Layer
compression
encoding
encryption
TLS, SSL
Session Layer
Session
Sockets
Transport Layer
01011 101
Segmentation
TCP, UDP
Reassembly
Network Layer
Packets
IP, ICMP, IPsec
Packets assembly
Data Link Layer
Frames
Ethernet, Wifi
Intra-network communications
Physical Layer
00100111
Sending cable
Fiber
Receiving cable

Working of OSI Model

➢ When a sender sends data:

1. Data moves from Layer 7 to Layer 1.
2. Each layer adds its own header information.
3. The data is transmitted through the network.
4. At the receiver side, data moves from Layer 1 to Layer 7.
5. Each layer removes its corresponding header and processes the data.

Explanation of Each Layer

1. Physical Layer (Layer 1)

- Responsible for transmitting raw bits (0s and 1s).
  - यह raw bits (0 और 1) को transmit करने के लिए responsible होती है।

- Defines cables, connectors, signals, voltage levels and transmission media.
  - यह cables, connectors, signals, voltage levels और transmission media को define करती है।

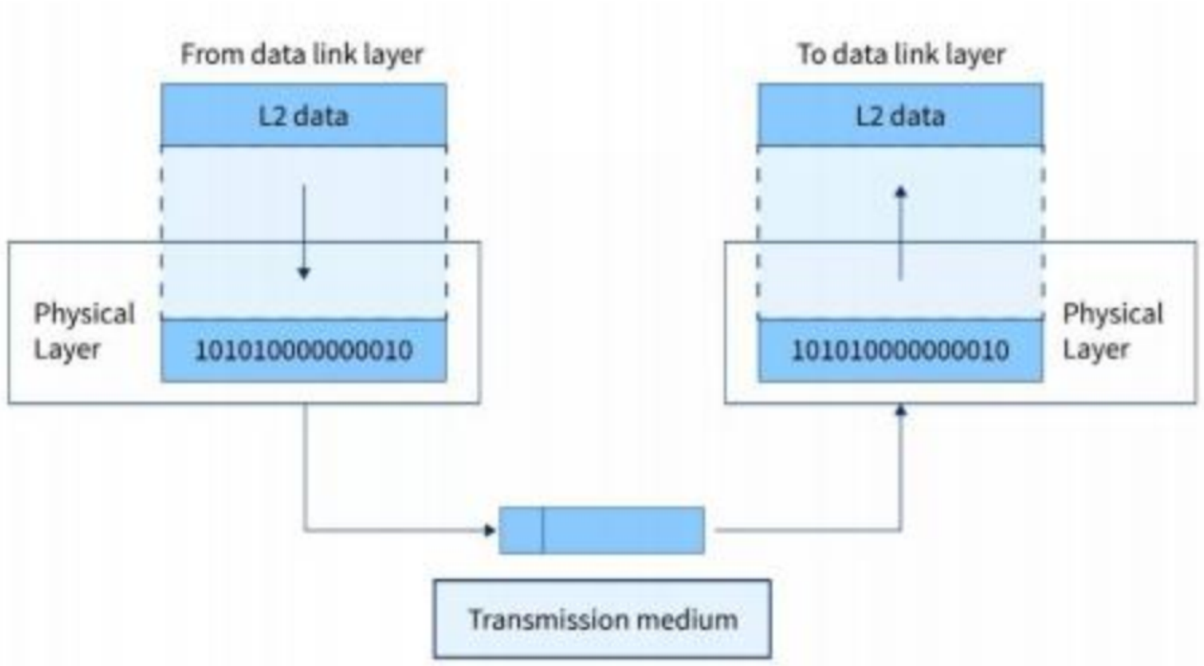
- Converts data into electrical, optical or radio signals.
  - यह data को electrical, optical या radio signals में convert करती है।

Devices Used

- Hub
- Repeater
- Cables
- Connectors

Example:

When a file is sent, the Physical Layer converts binary data into electrical signals and sends it through the cable.



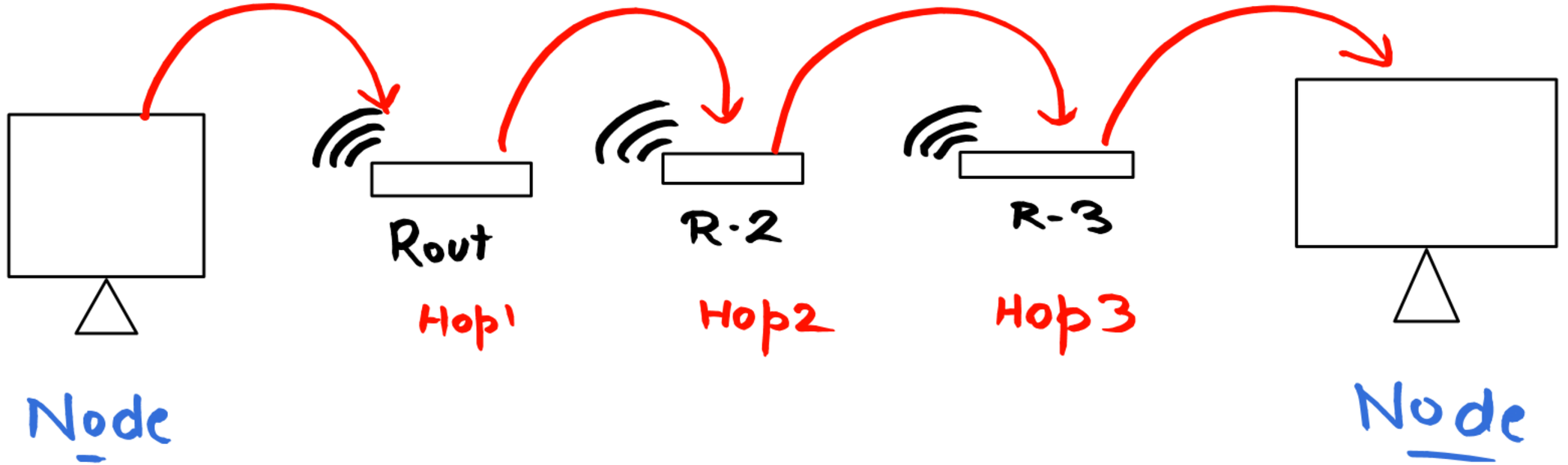
2.Data Link Layer (Layer 2)

MAC → Aadhar Card
IP → House Add.

- Provides node-to-node delivery.
  - यह node-to-node delivery प्रदान करती है।

- Detects and corrects transmission errors.
  - Transmission errors को detect और correct करती है।

- Uses MAC addresses for communication.
  - यह communication के लिए MAC addresses का उपयोग करती है।



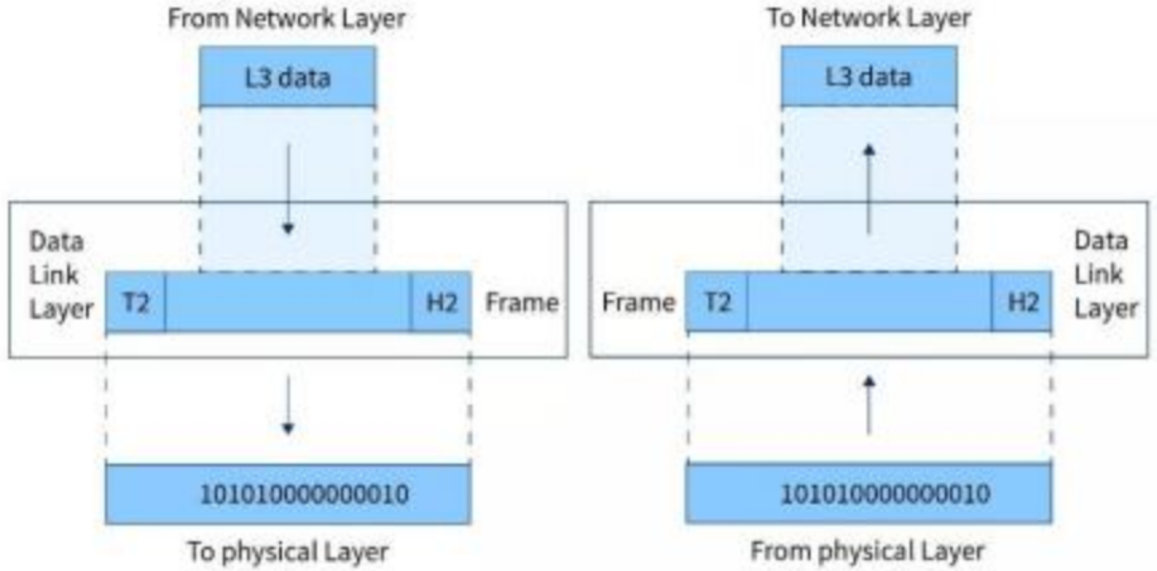
→ Node to Node

→ Hop to Hop

- Converts data into frames.
  - Data को frames में convert करती है।

Devices Used

- Switch
- Bridge



Example

A switch uses MAC addresses to forward data to the correct device.

CAM

3. Network Layer (Layer 3)

- Provides logical addressing using IP addresses.
  - यह IP addresses का उपयोग करके logical addressing प्रदान करती है।

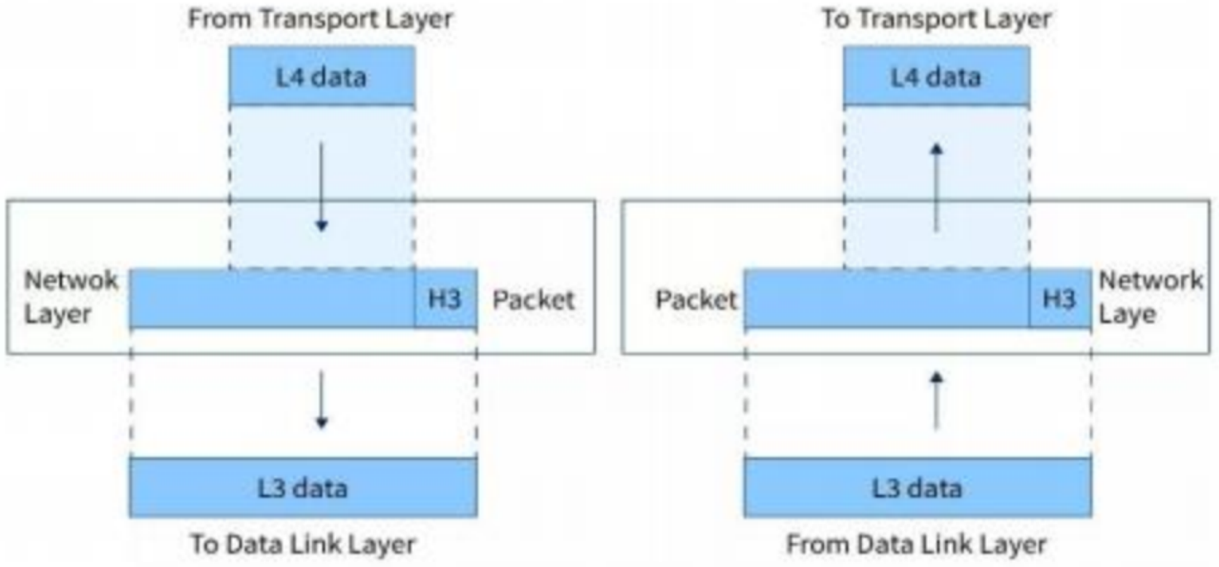
- Determines the best route for data transmission.
  - यह data transmission के लिए सबसे अच्छा route निर्धारित करती है।

- Performs routing between different networks.
  - यह अलग-अलग networks के बीच routing करती है।

- Converts data into packets.
  - Data को packets में convert करती है।

Devices Used

- Router



Example

When 4accessing a website, routers use IP addresses to find the destination server.

4. Transport Layer (Layer 4)

- **Provides end-to-end communication.**
  - End-to-end communication प्रदान करती है।

- **Ensures reliable data transfer.**
  - Reliable data transfer सुनिश्चित करती है।

- **Performs segmentation and reassembly.**
  - Data segmentation और reassembly करती है।

- Controls flow and error recovery.
  - Flow control और error recovery का कार्य करती है।

Protocols

- TCP
- UDP

Example

- TCP ensures that all parts of a downloaded file arrive correctly.

5. Session Layer (Layer 5)

- Establishes, manages and terminates communication sessions.
  - Communication sessions को establish, manage और terminate करती है।

- Synchronizes communication between devices.
  - Devices के बीच communication को synchronize करती है।

- Maintains active connections.
  - Active connections को बनाए रखती है।

Example

During a video conference, the Session Layer maintains the communication session.

6. Presentation Layer (Layer 6)

- Translates data into a common format.
  - Data को common format में translate करती है।

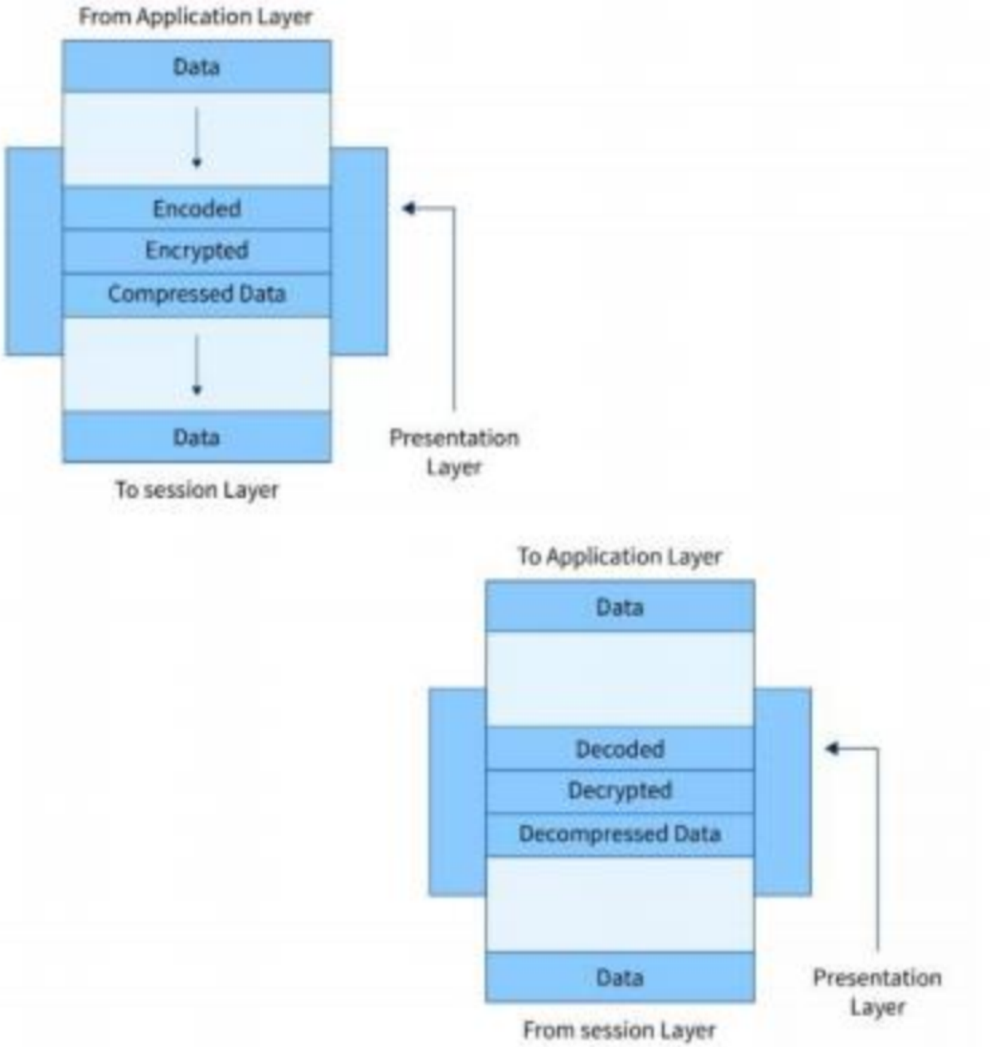
- Performs encryption and decryption.
  - Encryption और decryption करती है।

/encoding and decoding

- Performs data compression and decompression.
  - Data compression और decompression करती है।

Example

HTTPS uses encryption to secure data before transmission.



7. Application Layer (Layer 7)

- Provides services directly to end users.
  - यह सीधे end users को network services प्रदान करती है।

- Enables web browsing, email and file transfer.
  - Web browsing, email और file transfer को support करती है।

- Acts as an interface between user applications and the network.
  - यह user applications और network के बीच interface का कार्य करती है।

Protocols

- HTTP
- HTTPS
- FTP
- SMTP
- DNS

Example

When you open a website using a browser, the Application Layer provides HTTP services.

Advantages of OSI Model

1. Simplifies network communication.

Network communication को सरल बनाता है।

2. Standardizes networking functions.

Networking functions को standardize करता है।

3. Easy troubleshooting.

Troubleshooting आसान बनाता है।

4. Supports interoperability.

Interoperability को support करता है।

5. Modular architecture.

यह modular architecture प्रदान करता है।

Disadvantages of OSI Model

1. **Complex architecture.**

   इसका architecture जटिल (complex) है।

2. **Mostly theoretical.**

   मुख्यतः theoretical model है।

3. **Layer overhead increases processing.**

   Layer overhead processing बढाता है।

4. Some functions overlap.

कुछ functions एक-दूसरे से overlap करते हैं।

5. Implementation is costly.

Implementation महंगी हो सकती है।

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BOARD OF TECHNICAL EDUCATION
UTTAR PRADESH

DIPLOMA
3rd SEMESTER

COMPUTER SCIENCE & ENGINEERING

THANK YOU!

Stay Consistent, Keep Practicing and Always Believe in Yourself.

FOCUS	LEARN	PRACTICE	ACHIEVE
on Your Goal	Everyday	Consistently	Your Dreams

> "Your Hard Work Today
> Builds a Better Tomorrow."
> — All The Best! —

KEEP LEARNING, KEEP GROWING, KEEP ACHIEVING!